Budget-Constrained Confidence Ranking for Black-Box LLMs: Label-Free Top-k Selection with Sequential Sampling Guarantees

— Methodology and Experimental Setup (working draft v2) —

1 Methodology

1.1 Problem Formulation and Selection Optimality

Let f denote a black-box LLM accessible only through sampled responses: no token-level probabilities, no gradients, no training access. Let $\{x_i\}_{i=1}^{N}$ be a batch of binary questions, and let B be a total API-call budget for the batch. For case i and prompt phrasing j, let p_{ij} denote the probability that f emits case i's majority answer under phrasing j at fixed decoding parameters. Because confidence should not be an artifact of one phrasing, the target of inference is the *phrasing-marginal stability*

$$\bar{p}_i \;=\; \mathbb{E}_{J\sim Q}[p_{iJ}]\,, \tag{1}$$

where Q is a distribution over semantics-preserving paraphrases of x_i. The task is to select a subset $T \subset \{1,\dots,N\}$ with $|T| = k$ (here $k = 10$) maximizing the expected number of correct answers $\mathbb{E}[\#\text{correct in } T]$, subject to total spend at most B, *without labels*.

Every quantity we estimate is behavioral; a single explicitly stated assumption bridges behavior to correctness. Writing $\hat{Y}_i$ for the majority answer and $Z_i = \mathbf{1}\{\hat{Y}_i = Y_i\}$ for its correctness, the composite score s_i of Section 1.5 is a predictor of Z_i, and:

Assumption 1 (Monotonicity)**.** *The conditional accuracy $\pi(s) = \mathbb{P}(Z = 1 \mid s)$ is nondecreasing in s.*

Proposition 1 (Optimality of top-k selection)**.** *Under Assumption 1, the rule "select the k cases with the largest s_i" maximizes $\mathbb{E}[\#\text{correct in } T]$ over all size-k selection rules.*

Proof. By linearity of expectation (no cross-case independence is required), $\mathbb{E}[\#\text{correct in } T] = \sum_{i\in T}\pi(s_i)$. Let T be any size-k set that is not the top-k by s. Then there exist $a \notin T$ and $b \in T$ with $s_a > s_b$. The swap $T' = T \cup \{a\} \setminus \{b\}$ changes the objective by $\pi(s_a) - \pi(s_b) \geq 0$ by Assumption 1. Each such swap strictly increases the number of top-k members of the selected set, so after at most k swaps the top-k set is reached with the objective never decreased. Hence no size-k set attains a strictly larger objective. □

Remark 1. *Proposition 1 is the Neyman–Pearson exchange argument transplanted to selection: score plays the role of the likelihood ratio, the selection size k the role of the test size. It delivers optimality* given *the score and certifies that ranking-grade confidence is sufficient — no calibration ($\pi(s) = s$) is required, which is what permits a label-free pipeline. It is silent on the quality of the score itself, the subject of Sections 1.4–1.8. Assumption 1 is not verifiable without labels; Section 2 tests it empirically on labeled benchmarks, globally (score-decile shape) and locally at the top of the ranking, where the proposition consumes it.*

1.2 Budget-Optimal Sampling Design

Fresh-paraphrase sampling is exactly i.i.d. Repeated samples of a *single* phrasing are conditionally correlated through prompt-specific bias (wording effects, position effects), which degrades the information per call. Our protocol removes this by design.

Lemma 1 (Fresh-paraphrase i.i.d. sampling)**.** *Fix a case and suppress its index. Suppose each draw $t = 1, 2, \ldots$ is generated by the two-stage mechanism $J_t \sim Q$ i.i.d., then $Y_t \mid J_t \sim \mathrm{Bern}(p_{J_t})$, with the pairs (J_t, Y_t) independent across t. Then $Y_1, Y_2, \ldots$ are i.i.d. $\mathrm{Bern}(\bar{p})$ with $\bar{p} = \mathbb{E}_{J\sim Q}[p_J]$.*

Proof. Marginally, $\mathbb{P}(Y_t = 1) = \mathbb{E}_Q[p_{J_t}] = \bar{p}$ for every t, and independence of the pairs across t implies independence of the Y_t. A mixture of Bernoullis is a Bernoulli at the mixture mean. □

Lemma 1 motivates the protocol used throughout: *every response draw uses a freshly sampled paraphrase, one draw per paraphrase.* The cost of deviating from it is quantified by the classical cluster-sampling penalty:

Lemma 2 (Design effect of paraphrase reuse)**.** *Allocate n draws as g paraphrases $\times$ $m = n/g$ draws each, with per-draw marginal variance $\sigma^2 = \bar{p}(1-\bar{p})$ and within-paraphrase correlation $\rho \geq 0$. Then*

$$\mathrm{Var}\big(\hat{\bar{p}}\big) \;=\; \frac{\sigma^2}{n}\,[\,1 + (m-1)\,\rho\,]\,, \tag{2}$$

minimized over allocations at $m = 1$ (i.e. $g = n$), where the bracket equals 1.

Proof. $\mathrm{Var}(\sum_t Y_t) = n\sigma^2 + \sum_{t\neq t'} \mathrm{Cov}(Y_t, Y_{t'})$. Covariances vanish across paraphrases and equal $\rho\sigma^2$ within; there are $g\,m(m-1)$ ordered within-paraphrase pairs, giving $n\sigma^2[1 + (m-1)\rho]$. Divide by n^2. The bracket is increasing in m, hence minimized at $m = 1$. □

Lemma 1 is the $m = 1$ corner of Lemma 2 at which the i.i.d. description becomes exact rather than approximate; the *no-spread* ablation of Section 2.4 measures the bracket of Eq. (2) empirically. Two further consequences: the posterior of Section 1.3 is exact, and the estimand is unambiguously $\bar{p}$ of Eq. (1) *relative to the paraphrase frame Q*. The latter is the honest fine print: a paraphrase generator that drifts semantically does not violate independence but *contaminates the estimand*; we therefore audit meaning preservation on a development slice (Section 2.3).

Sequential triage via a truncated SPRT. Screening tests $H_0 : \bar{p} \leq p_0$ against $H_1 : \bar{p} \geq p_1$ (we use $p_0 = 0.7$, $p_1 = 0.95$) on the i.i.d. agreement stream of Lemma 1. After n draws with S_n agreements, Wald's sequential probability ratio test (Wald, 1945) tracks

$$\Lambda_n \;=\; S_n \log\frac{p_1}{p_0} \;+\; (n - S_n)\log\frac{1-p_1}{1-p_0} \;\approx\; 0.305\,S_n \;-\; 1.792\,(n - S_n), \tag{3}$$

stopping to accept H_1 when $\Lambda_n \geq A \approx \log\frac{1-\beta}{\alpha}$ and to accept H_0 when $\Lambda_n \leq B_\ell \approx \log\frac{\beta}{1-\alpha}$; at $\alpha = \beta = 0.1$, $A \approx 2.20$ and $B_\ell \approx -2.20$. By the Wald–Wolfowitz theorem (Wald and Wolfowitz, 1948), the SPRT minimizes the expected number of draws under both hypotheses among all tests with the same error rates. We truncate it into two stages:

- **Stage 1 (screen):** 3 draws; any disagreement ⇒ eliminate. Since $3 \times 0.305 = 0.92 < A$, stage 1 *never certifies*: unanimous survivors occupy the SPRT's continuation region.

- **Stage 2 (probe):** survivors receive 6 further draws (the continuation of the same stream), alongside the orthogonal probes of Section 1.4. Certification ($\Lambda \geq A$, i.e. ≈ 8 net-clean agreements) can only complete here.

The two-stage pipeline is thus a single truncated SPRT with its continuation region relabeled, not two ad hoc phases. Its economics are exactly computable:

Lemma 3 (Operating characteristics of the triage)**.** *(a) The stage-1 screen passes a case with probability exactly $\bar{p}^3$: 0.343 at the H_0 boundary ($\bar{p} = 0.7$) and 0.857 at the H_1 boundary ($\bar{p} = 0.95$). (b) Let $D(p) = p \log \frac{p_1}{p_0} + (1-p) \log \frac{1-p_1}{1-p_0}$ be the per-draw drift of Λ_n. Wald's identity $\mathbb{E}[\Lambda_N] = \mathbb{E}[N]\, D(p)$ gives, with the boundary approximations,*

$$\mathbb{E}[N \mid H_0] \approx \frac{\alpha A + (1-\alpha) B_\ell}{D(p_0)} \approx \frac{-1.76}{-0.324} \approx 5.4, \qquad \mathbb{E}[N \mid H_1] \approx \frac{(1-\beta) A + \beta B_\ell}{D(p_1)} \approx \frac{1.76}{0.200} \approx 8.8, \tag{4}$$

versus $n \approx 12$ for a fixed-sample test at the same error rates (normal approximation): sequential triage spends roughly 55% fewer draws on unstable cases and 27% fewer on stable ones.

Proof. (a) is the probability of three successes in three i.i.d. Bern($\bar{p}$) draws. (b) Λ_N is a stopped random walk with i.i.d. increments of mean $D(p)$; Wald's identity applies, and at stopping Λ_N sits approximately at a boundary, so $\mathbb{E}[\Lambda_N]$ equals the acceptance-probability-weighted boundary average; solve for $\mathbb{E}[N]$. The fixed-sample size follows from the two-sided normal approximation $n \geq \big(z_{1-\alpha}(\sigma_0 + \sigma_1)/(p_1 - p_0)\big)^2$. □

The step asymmetry in Eq. (3) — one disagreement (-1.79) outweighs nearly six agreements ($+0.305$ each), and drift is steeply negative under H_0 but shallow under H_1 — makes *rejection cheap and certification expensive*, which is precisely the economy a top-k funnel requires: most cases need only elimination-grade evidence. Stage-2 depth is fixed rather than adaptive so that the entropy estimates of Section 1.4 share comparable small-sample bias across cases; truncation of the joint two-stage procedure slightly perturbs the nominal error rates relative to Wald's bounds, quantified by simulation and reported with the implementation (stage 1 alone is exact by Lemma 3(a)).

1.3 Stability Posterior

Pooling a case's k_i i.i.d. draws with m_i agreements under the Jeffreys prior Beta($\frac{1}{2}, \frac{1}{2}$) yields the posterior

$$\bar{p}_i \mid \text{data} \;\sim\; \text{Beta}\big(m_i + \tfrac{1}{2},\; k_i - m_i + \tfrac{1}{2}\big)\,, \tag{5}$$

exact under Lemma 1. Two functionals are used downstream: the *lower credible bound* $L_i(\alpha) = F^{-1}_{\bar{p}_i|\text{data}}(\alpha)$, which is the first confidence signal and is budget-honest by construction (unanimity at small k_i yields only modest bounds), and the *tail mass*

$$\varepsilon_i \;=\; \mathbb{P}(\bar{p}_i < p_0^* \mid \text{data}) \;=\; F_{\text{Beta}(m_i + \frac{1}{2},\, k_i - m_i + \frac{1}{2})}(p_0^*) \tag{6}$$

at a stability standard p_0^* (we report $p_0^* = 0.8$), which feeds the selection guarantee of Section 1.6. For calibration of intuition (all values computed exactly): 9/9 unanimous draws give $\varepsilon \approx 0.042$; 12/12 give $\varepsilon \approx 0.019$; a single disagreement (8/9) gives $\varepsilon \approx 0.27$.

1.4 Signal Battery

Stage-2 survivors are scored on five signals, listed in the prior reliability order asserted by the cascade of Section 1.5 (an assertion that the Shapley analysis of Section 1.8 tests rather than presumes).

(1) Agreement lower bound. $L_i(\alpha)$ from Eq. (5).

(2) Invariance battery (specification test). Let g be an input transformation with a known action σ_g on the correct label: logical negation ($\sigma_g(y) = 1 - y$), candidate or option reordering (a relabeling), and insertion of an irrelevant distractor ($\sigma_g = \mathrm{id}$). Define the consistency indicator

$$C_i(g) \;=\; \mathbf{1}\{\, f(g(x_i)) = \sigma_g(f(x_i))\}\,, \tag{7}$$

and the signal as the pass fraction over the battery. Its necessity is an identifiability statement:

Lemma 4 (Power separation)**.** *Let H_{cue} denote the alternative under which the model's answer is a fixed function of a surface feature preserved by every paraphrase in Q, and H_{concept} the hypothesis that the model computes a function of the underlying proposition, answering correctly under label-action transformations up to error ϵ. Then: (i) the resampled answer stream $\{Y_t\}$ of Lemma 1 has the same law, $\mathrm{Bern}(\bar{p})^{\otimes n}$, under a stably wrong cue-follower as under a stably right concept-computer with the same $\bar{p}$; hence any test observing only $\{Y_t\}$ has power equal to its size against H_{cue}. (ii) A battery of r label-action transformations that destroy or invert the cue separates the hypotheses: the cue-follower passes all r with probability at most $(\frac{1}{2} + \delta)^r$ for binary answers, while the concept-computer passes with probability at least $(1 - \epsilon)^r$.*

Proof. (i) Both hypotheses induce i.i.d. $\mathrm{Bern}(\bar{p})$ draws with the same $\bar{p}$; identical laws yield identical rejection probabilities for any measurable test, so power cannot exceed size. (ii) Under H_{cue} the transformed input carries a destroyed or inverted cue, making agreement with $\sigma_g(f(x))$ a $\approx \frac{1}{2}$-probability event per probe; independence across probes gives the product bounds. □

Lemma 4(i) is why no amount of additional sampling substitutes for the battery: resampling explores the model's randomness at a fixed input, while the probes explore the model's *function* across inputs with known label action. Its contrapositive delimits the method's blind spot — a cue invariant under Q *and* under every g in the battery defeats both instruments — which we state in the limitations. The battery is the component aimed at the stable-but-wrong regime.

(3) Semantic entropy over rationales. The k_i explanations accompanying the sampled answers are partitioned into rationale-equivalence clusters $c = 1, \ldots, \hat{M}_i$ by a cross-provider LLM judge ("do these two explanations give the same reason?"); with $\hat{q}_c$ the fraction of explanations in cluster c, we use the Miller–Madow bias-corrected entropy (Miller, 1955)

$$\widehat{\mathrm{SE}}_i \;=\; -\sum_c \hat{q}_c \log \hat{q}_c \;+\; \frac{\hat{M}_i - 1}{2k_i}\,. \tag{8}$$

Answer-level agreement is a statistic on $\{0, 1\}$ and saturates at unanimity; $\widehat{\mathrm{SE}}_i$ is a statistic on the far larger rationale space and remains informative exactly where agreement is degenerate (Kuhn et al., 2023; Farquhar et al., 2024): unanimity reached through one consistent argument outranks unanimity reached through several incompatible arguments. The correction term removes the leading-order plug-in bias, which depends on the cluster count $\hat{M}_i$ and would otherwise differ across cases even at equal depth; equal stage-2 depth k_i keeps the residual higher-order bias comparable.

(4) Challenge resistance. The model is shown its own answer together with a standardized counterargument and asked to reconsider; the signal is whether it holds or capitulates. Because preference-trained models capitulate partly as a trained reflex (Sharma et al., 2024), the signal is used *comparatively within a model*: cases that survive challenge outrank cases that do not.

(5) Verbalized confidence. A temperature-0 self-reported confidence. Preference optimization inflates and clusters stated confidences while largely preserving their rank information (Tian et al., 2023; Xiong et al., 2024); the signal is therefore admitted as a final tie-breaker only, never load-bearing.

1.5 Aggregation

The primary combiner is a *lexicographic cascade*: unanimity filter → invariance pass ($\geq 2/3$ of the battery) → rank by $\widehat{\mathrm{SE}}_i$ (ascending) → challenge resistance → verbalized confidence. The cascade is Borda aggregation with infinitely separated weights; it encodes the asserted reliability ordering and yields an interpretable rejection reason at every stage. As a robustness alternative we report *Borda (average-rank) aggregation* over the five signals, the assumption-minimal combiner: ranks are the maximal invariants under monotone transformations of each signal (no implicit scale weighting), the Borda count is consistent under Mallows-type ranking noise (Fligner and Verducci, 1986), and average rank minimizes the total Spearman-footrule distance to the input rankings, a 2-approximation to the NP-hard Kemeny-optimal aggregate (Diaconis and Graham, 1977). Material disagreement between the two combiners is reported and adjudicated by the Shapley values of Section 1.8.

1.6 Label-Free Selection Guarantee

Proposition 2 (Selection guarantee)**.** *Let T be any selected set and ε_i the posterior tail masses of Eq. (6). Then, with no independence assumption across cases,*

$$\mathbb{E}[\,\#\{\,i \in T : \bar{p}_i < p_0^*\,\} \mid \text{data}] \;=\; \sum_{i \in T} \varepsilon_i, \qquad \mathbb{P}(\exists\, i \in T : \bar{p}_i < p_0^* \mid \text{data}) \;\leq\; \sum_{i \in T} \varepsilon_i. \tag{9}$$

Proof. Write $\#\{i \in T : \bar{p}_i < p_0^*\} = \sum_{i \in T} \mathbf{1}\{\bar{p}_i < p_0^*\}$ and take conditional expectations term by term; linearity requires no independence. The second statement is the union bound over the same events. □

This is a Bayesian FDR-style control on the expected number of sub-standard cases *in the delivered set*: if every selected case is 12/12 unanimous, the expected number with true stability below 0.8 is at most $10 \times 0.019 \approx 0.2$. The guarantee inverts into a design equation:

Corollary 1 (Certification depth)**.** *Under a uniform prior, unanimity in k draws gives the closed form $\varepsilon = (p_0^*)^{k+1}$; hence $\varepsilon \leq \varepsilon_0$ requires $k \geq \log \varepsilon_0 / \log p_0^* - 1$. The Jeffreys posterior, with its mass concentrated toward 1, is numerically tighter: at $p_0^* = 0.8$, the minimal unanimous depth is $k = 9$ for $\varepsilon_0 = 0.05$ and $k = 12$ for $\varepsilon_0 = 0.02$ (versus 13 and 17 from the uniform-prior bound).*

Proof. With a uniform prior and k successes in k draws the posterior is $\mathrm{Beta}(k+1, 1)$, whose CDF at p_0^* is $(p_0^*)^{k+1}$; invert. The Jeffreys values are exact evaluations of Eq. (6). □

The closed form reads directly: $(p_0^*)^{k+1}$ is the probability that a case no more stable than the standard could nevertheless have produced the observed unbroken streak, so residual doubt decays geometrically in streak length at rate p_0^* — certifying against a higher standard is exponentially more expensive. Corollary 1 is how stage-2 depth is *derived*: $k = 9$ is the minimal certification depth for the reported guarantee anchor $(\varepsilon_0, p_0^*) = (0.05,\, 0.8)$ (Section 2.3).

Eq. (9) conditions on all observed data, selection included, and is exact as a posterior statement; its frequentist calibration under selection is a separate question, which we flag rather than claim.

The division of epistemic labor is deliberate: *stability control* (Proposition 2) and *selection optimality given the score* (Proposition 1) are proven; the stability-to-correctness bridge (Assumption 1) is assumed, stated once, and tested empirically in Section 2.

1.7 Evidence Exchange: Converting Probes into Effective Draws

Draws and probes are heterogeneous evidence with a common currency: the log-likelihood ratio. The SPRT prices an agreement at $\log(p_1/p_0) \approx 0.305$ nats; by Lemma 4, a label-action probe passed under H_{concept} versus H_{cue} contributes $\log(q_{1,g}/q_{0,g})$ nats, where $q_{1,g} = 1 - \epsilon$ and $q_{0,g}$ is the cue-follower's pass probability. Define the *exchange rate* of probe g:

$$\kappa_g \;=\; \frac{\log(q_{1,g}/q_{0,g})}{\log(p_1/p_0)} \qquad \text{(unanimous draws per passed probe).} \tag{10}$$

At $\epsilon = 0.05$: a passed negation or candidate-swap probe ($q_0 = \frac{1}{2}$) is worth $\kappa \approx 2.1$ draws; a passed distractor probe ($q_0 \approx 0.8$, since an inserted distractor leaves most surface cues intact) is worth $\kappa \approx 0.56$; a *failed* label-action probe contributes $\log(\epsilon/\frac{1}{2}) \approx -7.5$ agreement-units. The theory thus ranks probe types by design rather than by fiat, and prices probe failure with the same productive asymmetry as the SPRT itself.

Proposition 3 (Evidence exchange)**.** *Assume the composite-evidence model: conditional on the case's latent status (stable-and-concept-driven versus not), draw outcomes and probe outcomes are jointly independent with the margins above. Then total evidence adds, $\Lambda^{\text{tot}} = \Lambda^{\text{draws}} + \sum_g \Lambda_g^{\text{probe}}$, and a case with m unanimous draws and passed probe set G carries certification evidence equivalent to*

$$k_{\text{eff}} \;=\; m + \sum_{g \in G} \kappa_g \tag{11}$$

unanimous draws.

Proof. Log-likelihood ratios of conditionally independent observations add; dividing the probe terms by the per-draw rate $\log(p_1/p_0)$ expresses the sum in draw units. □

Illustratively, with the deployed battery ($\sum_g \kappa_g \approx 4.8$): 9 draws + battery yield $k_{\text{eff}} \approx 13.8$ and $\varepsilon_{\text{eff}} \approx 0.012$ at 12 calls — more evidence than 12 draws alone ($\varepsilon \approx 0.019$) at the same spend — and 6 draws + battery (9 calls) match ≈ 11 draws (11 calls): the same confidence roughly 20% cheaper. Per call, a label-action probe at $\kappa \approx 2.1$ strictly dominates a draw, upgrading Lemma 4's qualitative conclusion into an allocation rule and giving the Shapley analysis of Section 1.8 a prediction to test rather than only numbers to report.

Two caveats scope the result honestly. First, Proposition 3 is *model-based*: draws evidence stability, probes evidence mechanism, and summing them presumes a composite latent status of which both are symptoms — ε_{eff} is a posterior statement under that model, whereas Proposition 2 remains the exact, assumption-free stability guarantee on draws alone; we layer the two rather than replace one with the other. Second, conditional independence is plausibly violated (probes and draws share model and case), and correlated evidence double-counts, making Eq. (10) an upper bound on the honest rate; we therefore (i) report results at κ and $\kappa/2$, and (ii) estimate the empirical rate $\hat{\kappa}$ on labeled benchmarks by logistic regression of correctness on draw count and probe passes, testing the derived rate directly (Section 2.5). The evidence-equivalence experiment of Section 2.4 pre-registers the resulting prediction. (An anytime-valid formalization via e-values, in which independent evidence multiplies and the exchange is exact in log-e units, is available (Ramdas et al., 2023); we retain the likelihood-ratio presentation for accessibility.)

1.8 Design-Time Signal Valuation via Exact Shapley Analysis

The cascade ordering and the stage-2 probe allocation are design choices; on labeled benchmarks (where labels are legitimate at design time) we replace judgment with measurement. Define a cooperative game whose players are the five signals $\mathcal{S}$, and for $C \subseteq \mathcal{S}$ let $V(C)$ be the batch-averaged prec@10 of the pipeline restricted to the signals in C (Borda-combined), with $V(\emptyset)$ the random-selection reference. The Shapley value of signal j is

$$\phi_j \;=\; \sum_{C \subseteq \mathcal{S}\setminus\{j\}} \frac{|C|!\;(|\mathcal{S}|-|C|-1)!}{|\mathcal{S}|!}\,[V(C\cup\{j\})-V(C)]\,. \tag{12}$$

With $|\mathcal{S}| = 5$ there are $2^5 = 32$ coalitions, each a re-scoring pass over cached samples, so Eq. (12) is computed *exactly* — the mirror image of the data-valuation setting of Ghorbani and Zou (2019), where thousands of players force Monte Carlo approximation. Three properties are used: efficiency, $\sum_j \phi_j = V(\mathcal{S}) - V(\emptyset)$, partitions the total gain over random selection across signals; subset averaging assigns fair credit to *redundant* signals (e.g. agreement and rationale entropy, both driven by sampling consistency), where leave-one-out ablation assigns them zero (Ghorbani and Zou, 2019); and the cost-normalized values ϕ_j/c_j, with c_j the API calls signal j consumes, justify the probe-budget allocation and are compared against the derived exchange rates of Eq. (10) — theory and measurement of the same allocation question. The coalition contrast $V(\{L\})$ versus $V(\{L, \text{flip}\})$ additionally provides a direct measurement of the stable-but-wrong recovery claimed for the invariance battery. Shapley valuation is a design-time, benchmark-only analysis; deployment runs label-free.

1.9 Deployment Loop

In deployment the pipeline runs on recurring batches with no labels. Assumption 1 is monitored through label-free surrogates: survival on a held-out perturbation family never used in selection, and agreement with an independent model family, both of which should be nondecreasing in score decile if the ranking retains meaning. The selected top-k of each batch is human spot-checked together with blinded non-selected controls; accumulated labels graduate the system from ranking-grade use (top-k selection, the present paper) toward threshold-grade use (conformal routing) and probability-grade use (calibrated risk inputs).

2 Experimental Setup

The experiments are organized as a proof structure. The value claim — *confidence measurement is valuable under a limited call budget* — decomposes into: (C1) measuring at all beats not measuring; (C2) behavioral measurement beats self-report; (C3) how a fixed budget is *allocated* matters — the paper's own claim; (C4) the value generalizes across failure regimes and models, with known boundaries. Each element below instantiates or falsifies exactly one methodology component and carries exactly one sub-claim; Table 1 states the mapping.

2.1 Datasets and Task Construction

A confidence method's value depends on *why* the model errs, so each canonical error cause receives one arena; all three datasets sit in the standard UQ evaluation canon (Kuhn et al., 2023; Farquhar et al., 2024; Tian et al., 2023; Vashurin et al., 2025), tying our numbers to published yardsticks.

- **TriviaQA-verification** (Joshi et al., 2017) — *wrong memorized facts*: the stable-but-wrong regime, the known blind spot of agreement-based confidence (Lemma 4(i)) and the arena in which the invariance battery must earn its place; the coalition contrast $V(\{L\})$ vs. $V(\{L, \text{flip}\})$ is measured here. Items are converted to verification format, "*Is [candidate] the correct answer to [question]?*", with candidates drawn 50/50 from gold answers and hard distractors (the model's own wrong greedy answers on a third, disjoint item pool — disjoint from both the evaluation and development slices, so that distractor construction leaks no evaluation labels), yielding realistic difficulty and balanced labels.
- **StrategyQA** (Geva et al., 2021) — *broken reasoning chains*: natively binary, implicit multi-step reasoning; the regime where answer agreement saturates while rationale structure still discriminates, i.e. where semantic entropy (Eq. (8)) carries the load.
- **MMLU** (Hendrycks et al., 2021) — *knowledge boundaries; the adversarial pick*: for black-box instruction-tuned models, verbalized methods have been reported to outperform sampling-based methods on MMLU (Vashurin et al., 2025), so this dataset is included as a planned stress test of the cascade ordering and a pre-registered candidate locus of Monotonicity weakening. Items are binarized to the same verification format (option-order manipulation remains available as a perturbation), keeping one uniform pipeline across datasets.

The verification construction engineers base accuracy into a middle band: at base accuracy near 1, prec@10 is uninformative for any selector (the hypergeometric null under random selection makes this precise), and at low base accuracy high-confidence cases barely exist; selection lift is measurable only between. Per dataset: 500 evaluation items (fixed seed) partitioned into 10 disjoint batches of 50 with top-10 selected per batch — matching the selection objective of Section 1.1, supplying 10 paired observations of prec@10 per configuration, and mirroring the recurring-batch deployment shape of Section 1.9 — plus 100 development items on which every hyperparameter (temperature, SPRT thresholds, paraphrase and judge prompts, cascade thresholds, budget-level allocation rules) is frozen before any evaluation batch is touched. Nothing evaluative touches the development slice and nothing tuning touches the 500 evaluation items; the selection pipeline itself is label-blind everywhere by construction, labels entering only at metric computation after all selections are frozen.

2.2 Models and Judge

Three model families spanning capability tiers and providers: one frontier API model, one mid-tier model from a different provider, and one open-weights model served via a hosted API (the mid-tier arm is the most deployment-relevant: weaker models carry more genuine uncertainty, hence more selectable signal, matching the budget-constrained deployments the method targets). The rationale-clustering judge is always a mid-tier model from a *different* provider than the model under test, rotated accordingly (same-model judging is a correlated-error confound); a judge-sensitivity check swaps in a cheaper judge on one dataset. For the label-free audit, each model is cross-checked by the open-weights model, and the open-weights model by the mid-tier model.

2.3 Pipeline Configuration

Per case, a pool of 12 paraphrases is generated offline by an LLM with a fixed prompt and audited for meaning preservation on the development slice (the estimand check of Section 1.2). Every response draw uses a fresh paraphrase, one draw each (Lemma 1 operationalized). Stage 1: 3 paraphrases $\times$ 1 draw at temperature 0.7; any disagreement eliminates (operating characteristic exactly $\bar{p}^3$, Lemma 3(a)). Stage 2 (survivors): 6 further fresh paraphrases $\times$ 1 draw — total $k_i = 9$

i.i.d. draws feeding Eq. (5), the minimal certification depth for the reported guarantee anchor $(\varepsilon_0, p_0^*) = (0.05, 0.8)$ by Corollary 1; 3 invariance probes at temperature 0 (negation, candidate swap, distractor insertion); 1 challenge probe; 1 verbalized-confidence call at temperature 0; 1 judge call clustering the 9 rationales. A fourth perturbation family, *entity renaming*, is generated offline for all cases and reserved exclusively for the label-free audit — never used in selection — and *executed* only where the audit compares, on per-batch ranks 1–30 (300 calls per dataset–model configuration). Expected pipeline cost is $3 + (\text{survival rate}) \times 12$ calls per case (≈ 8 at 40% survival). Every raw response is cached (prompt, parameters, output, content hash), so all baselines, ablations, budget curves, and Shapley coalitions below are re-scoring passes over the cache at zero marginal API cost.

2.4 Baselines, Ablations, and the Evidence-Equivalence Arms

All comparisons are at *matched total batch budget* (the pipeline's per-case spend is adaptive, so budgets are matched in expectation at the batch level). Baselines: (i) random selection, with the exact hypergeometric null as reference; (ii) verbalized confidence only, reported both at matched budget and at its native one-call cost; (iii) naive self-consistency (Wang et al., 2023): 8 draws of one fixed phrasing, ranked by agreement — the critical comparison, since it isolates allocation (C3) at identical spend; (iv) semantic entropy alone (Kuhn et al., 2023): 8 same-prompt draws plus clustering; (v) BSDetector (Chen and Mueller, 2024), the closest prior combined score, reconstructed from cached samples where its components permit. Ablations of the full method: *no-spread* (9 draws over 3 phrasings — deliberately reintroducing the design effect, the empirical measurement of the bracket in Eq. (2)); *no-SPRT* (flat 8 calls per case, no triage; expected to recover the cost ratios of Lemma 3(b)); *no-flip* (battery removed). The ablations feed Eq. (12) directly.

Evidence-equivalence arms. Proposition 3 pre-registers a quantitative prediction, tested with three budget-annotated arms constructed from cache: (A) 12 draws only (12 calls, $k_{\text{eff}} = 12$); (B) 6 draws + the probe battery (9 calls, $k_{\text{eff}} \approx 10.8$ at the derived rates); (C) 9 draws only (9 calls, $k_{\text{eff}} = 9$). The prediction is prec@10(B) $\approx$ prec@10(A) $>$ prec@10(C): arm B buys approximately arm A's quality at arm C's price. If B falls between A and C, the gap measures the evidence double-counting flagged in Section 1.7, and the empirically fitted $\hat{\kappa}$ replaces the derived rate; either outcome is informative.

2.5 Evaluation Protocol

Primary: the distribution of prec@10 over the 10 batches, paired per batch against every baseline (Wilcoxon signed-rank across batches; batch pairing absorbs batch-difficulty variance), plus a pooled Jeffreys interval over the 100 selected cases per configuration; lift over the exact per-batch hypergeometric null is reported both absolutely and as relative error reduction. **Standard:** AUROC and PRR (Vashurin et al., 2025) over all 500 items, and risk–coverage curves — the numbers directly comparable to published black-box UQ results. Both require a total order over all 500 cases: stage-1-eliminated cases rank below all survivors, ordered among themselves by stage-1 agreement pattern with random tie breaking, and cascade strata are handled by mid-rank ties (a tied correct–incorrect pair contributes $\frac{1}{2}$ to AUROC). **Headline figure:** prec@10 and AUROC as a function of calls per case ($3, 6, 9, 12$, subsampled from cache; each subsample repeated 20 times with distinct seeds and averaged, with error bands bootstrapped over batches, the real unit of uncertainty) for the full method versus naive self-consistency versus verbalized confidence — the budget claim (C3) as a picture; the low-budget vertical gap *is* the demonstrated value of designed measurement under constraint, and the horizontal gap reads as the calls self-consistency needs to

match the designed pipeline. **Assumption tests:** accuracy by score stratum (ten equal-count rank strata of 50) with per-stratum Jeffreys intervals and an isotonic-versus-unconstrained fit comparison (global Monotonicity); accuracy of per-batch ranks 1–10 vs. 11–30 vs. 31–50, pooled over the 10 batches, by Mann–Whitney (local Monotonicity, where Proposition 1 consumes it). **Robustness:** Jaccard overlap of the top-10 under fully re-drawn stage-2 sampling on two batches per dataset, reported against the chance floor $J \approx 0.11$ for two random 10-of-50 selections (selection must be a property of cases, not sampling noise); survival on the held-out entity-renaming family, selected (ranks 1–10) versus matched controls (ranks 11–30), one-sided — the label-free surrogate of Section 1.9, on probes the selector never saw; judge-swap sensitivity as (a) Spearman correlation of $\widehat{\mathrm{SE}}$ and (b) Jaccard of the resulting top-10s, the decision-level quantity; three model families throughout. **Design analysis:** the exact Shapley table of Eq. (12) per dataset and model, with the efficiency identity as a check row; the cost-normalized ϕ_j/c_j table (entropy attributed its judge call only, since it reuses the agreement draws — stated as a convention) compared against the derived exchange rates κ_g of Eq. (10); the empirical exchange rate $\hat{\kappa}$ from logistic regression of correctness on draw count and probe passes; and the $V(\{L\})$ vs. $V(\{L, \mathrm{flip}\})$ contrast quantifying stable-but-wrong recovery. **Reporting discipline:** the single confirmatory, Holm-corrected family is the pre-registered comparison of the full method against matched-budget self-consistency per dataset–model; all other results are reported as estimates with intervals. Every dataset–model cell appears in the tables, including those where a baseline wins (MMLU–verbalized is expected to be competitive). All figures and tables regenerate from the cache by a single script, and the development-slice freeze date is stated.

2.6 Design-to-Methodology Mapping

Table 1 summarizes the correspondence. Read as a whole: the datasets give each canonical error cause one arena (including the one where published evidence predicts our approach is weakest); the baseline ladder isolates, at matched budget, the value of measuring, of measuring behaviorally, and of allocating by design; the evidence-equivalence arms test the derived exchange rate between sampling and probing; and the quality-versus-spend figure states the limited-budget value claim as a single visual gap.

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Design element	Methodology	Claim carried / tested
TriviaQA-verification	Lemma 4, §1.4 (2)	Stable-but-wrong regime; invariance battery earns its place; $V(\{L\})$ vs. $V(\{L, \text{flip}\})$.
Disjoint distractor pool	§2.1	Distractor construction leaks no evaluation labels.
StrategyQA	§1.4 (3)	Agreement saturation; rationale entropy discriminates beyond agreement.
MMLU (adversarial)	§1.5, Ass. 1	Pre-registered hard case: cascade ordering under stress; candidate Monotonicity boundary.
Verification format, 50/50 hard distractors	§2.1	Base accuracy in the band where selection lift is measurable (hypergeometric null).
10 batches × 50, top-10	§1.1, §1.9	Evaluated object = the selection rule; inferential power for a 10-Bernoulli statistic; deployment shape.
Fresh paraphrase per draw	Lemma 1	Exact i.i.d.; posterior of Eq. (5) exact.
3-then-6 draws, eliminate on disagreement	§1.2, Lemma 3	Truncated SPRT; screen characteristic $\bar{p}^3$ exact; certification completes in stage 2.
Stage-2 depth $k = 9$	Cor. 1	Derived minimal depth for the guarantee anchor $(0.05, 0.8)$ — design as a theorem's output.
Dev-slice paraphrase audit	§1.2	Estimand integrity of $\bar{p}$ w.r.t. frame Q.
Fixed stage-2 depth; MM-corrected entropy	§1.4 (3)	Comparable entropy bias across cases; leading-order bias removed.
Random baseline	C1	Value of measuring at all, against the exact hypergeometric null.
Verbalized-only baseline	C2, §1.4 (5)	Behavioral beats self-report; RLHF-miscalibration rationale.
Self-consistency at matched 8 calls	C3, §1.2	*Core claim*: identical spend, different allocation — design carries the budget-constrained value.
No-spread / no-SPRT / no-flip ablations	Eq. (2), Lemma 3, Lemma 4	Componentwise price of the design; inputs to Eq. (12).
Evidence-equivalence arms A/B/C	Prop. 3	Pre-registered prediction B ≈ A > C; gap measures evidence double-counting; $\hat{\kappa}$ vs. κ.
Quality-vs-calls figure	§1.2	C3 as a picture; value at low budget is the vertical gap.
Stratum / isotonic / top-strata tests	Ass. 1, Prop. 1	Global and local Monotonicity (per-batch strata 1–10 / 11–30 / 31–50, pooled), where the optimality proof consumes it.
Jaccard under re-seeding (vs. chance floor 0.11)	§2.5	Selection is a property of cases, not sampling noise.
Held-out entity renaming (executed on ranks 1–30)	§1.9	Label-free audit on evidence the selector never saw (non-circular; matched controls).
Cross-provider judge + judge swap	§1.4 (3)	Bounds sensitivity of the one non-estimator component, at signal and decision level.
Exact Shapley over 32 coalitions	§1.8	Audits cascade ordering; ϕ_j/c_j vs. κ_g: measured and derived allocation compared.
Response caching	§2.3	All comparisons as zero-cost re-scores of one collection.

Table 1: Mapping from experimental design elements to methodology components and the specific